

Ramakrishna Jupalli

Senior Data Engineer

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PROFESSIONAL SUMMARY

Data & Software Engineering Leader with over 10 years of experience in designing and implementing scalable data pipelines, optimizing cloud-based architectures, leveraging advanced analytics to drive business insights and architecting large scale data solutions across retail, energy, and IoT domains.

- Proficient in **Python, R and PowerBI** for data manipulation, analysis, and visualization, utilizing libraries such as **Pandas, NumPy, Matplotlib**, and **Seaborn**
- Demonstrated expertise in building and optimizing **data warehousing** solutions, particularly with **Snowflake**, and a strong command of writing complex SQL queries for data retrieval and analysis.
- Applied advanced **statistical techniques** to analyze complex datasets, identify patterns, and extract actionable insights, facilitating data-driven decision-making.
- Developed **predictive** models using **machine learning** techniques, including **regression, classification**, and **clustering**, to forecast future outcomes.
- Leveraged cloud services like **AWS SageMaker** and **Google AI Platform** for streamlined Machine learning model development and deployment.
- Strong grasp of data integration concepts, **ETL/ELT** methodologies, and pipeline optimization, with experience using tools like **Talend** and **Informatica**.
- Proficient in setting up and managing **CI/CD pipelines using Jenkins, Github Actions** automating build, test, and deployment processes to ensure efficient software delivery.
- Skilled in designing and **executing A/B tests** to evaluate the impact of changes, optimizing user experiences, and driving business growth.
- Automated model testing and evaluation, integrating continuous integration and continuous deployment (**CI/CD**) processes into the machine learning workflows.
- Implemented containerization with **Docker** and orchestration with **Kubernetes** to deploy machine learning models in production environments.
- Experienced in building **Snow pipe, StreamLit** and In-depth knowledge of Data Ingestion, Sharing in Snowflake and Snowflake Database, Schema and Table structures.
- Designed and developed logical and physical data models that utilize concepts such as **Star Schema**, Snowflake Schema and **Slowly Changing Dimensions**.
- Experienced in working within **Agile/Scrum** environments, ensuring efficient

- project delivery and continuous improvement using **Atlassian** products
- Knowledge on **Google Cloud Platform (GCP)** services like compute engine, cloud load balancing, cloud storage, cloud Dataproc, Cloud Pub/Sub, cloud SQL, **Big Query**, stack driver monitoring, Cloud spanner, **looker**, and deployment manager
- Hands-on experience in handling database issues and connections with **SQL and No SQL** databases such as **MongoDB, HBase, DynamoDb, Redis, PostgreSQL, SQLite, SQL server**
- Hands-on experience on AWS components such as EMR, EC2, S3, RDS, IAM, Auto Scaling, CloudWatch, to ensure a secure zone for an organization in AWS public cloud

TECHNICAL SKILLS

- **Data Visualization & Reporting Tools::** Tableau, Power BI, SQL Server Reporting Services (SSRS), Microsoft Business Intelligence (SSIS, SSAS, SSRS), Looker, Datawrapper
- **Programming Languages & Libraries::** C, C++, SQL, R Programming Language, R Studio, Python (Pandas, NumPy, Matplotlib, Seaborn), R (tidyverse, dplyr, tidyr, ggplot2), Java, YAML, XML, JSON, ProtoBuf, JavaScript, Scala, Spark SQL, T-SQL (Transact-SQL), NLTK, PySpark, Apache Kafka, Splunk, Scikit-learn, TensorFlow, PyTorch, MQTT
- **Big Data & Analytics Platforms:** ADLS (Azure Data Lake Storage), ADF (Azure Data Factory), Azure Data Warehouse, Azure Synapse Analytics (formerly known as Azure SQL Data Warehouse), Azure Databricks HDInsight (Azure's cloud-based big data service), Apache park, Hbase, Apache Sqoop, Apache Pig, MapReduce
- **Database Systems:** MySQL, SQL Server, Databricks, Snowflake, Redis, PostgreSQL, SQLite, Oracle DB, Mongo DB, Amazon RDS (Relational Database Service), Oracle Exadata, Amazon DynamoDB
- **Cloud Platforms: Amazon Web Services (AWS):** AWS IoT, Amazon Relational Database Service (RDS), AWS Load balancer Redshift, Elastic MapReduce (EMR), S3, AWS Lambda, Quicksight, Boto3, Sage Maker, EC2, CloudWatch, CloudFront, Route 53, Athena
Microsoft Azure: Cloud Services (PaaS & IaaS), Active Directory, Application Insights, Azure Monitoring, Data Lake, Data factory, Azure Synapse Analytics (DW), Databricks, SQL Datawarehouse, **Google Cloud:** BigQuery, Vertex AI, Tensorflow on GCP, Google Cloud Storage, DataFlow, AutoML, Pub/Sub
- **OS:** Linux, Windows, MacOS, Yocto, OpenWRT, Zephyr

EDUCATION

MASTER OF SCIENCE (M.Sc.) IN ELECTRICAL ENGINEERING

University of Houston, Houston, TX

BACHELOR OF ENGINEERING (B.ENG.) (HONS.) IN ELECTRONICS AND COMMUNICATION ENGINEERING

Osmania University, Hyderabad, India

EXPERIENCE AND IMPACT

SUNPOWER CORPORATION, MCLEAN, VA, JAN 2021 TO PRESENT

Lead Data Engineer

- Designed and implemented robust ETL pipelines to integrate data from distributed IoT devices into centralized repositories, enabling real-time monitoring and analysis
- Orchestrated data extraction and transformation using **SQL Stored procedures**, python scripts and automated report and dashboard generation in **Power BI** for accurate insights and enhanced operational performance
- Developed automated workflows using tools like **Airflow** and scripting in **Python**, reducing manual data processing time by **50%**
- Conducted extensive performance tuning on **Kafka** and **Redshift clusters**, including table partitioning, indexing, and query optimization, resulting in a **25% improvement** in query performance and reduced load
- Employed **PySpark** for large-scale data processing and transformation, improving the speed and efficiency of handling **big data volumes**
- Scaled pipeline architecture to support a **50% increase** in data volume due to expanding device integration and user base growth
- Developed detailed documentation in **Confluence** and **Github** and implemented best practices for data pipeline maintenance, ensuring team-wide adherence to standardized processes
- Deployed monitoring and alerting systems with **CloudWatch** and **Datadog** to track pipeline health and address failures proactively
- Partnered with product and analytics teams to design data models and provide scalable solutions, leading to a **20% improvement in reporting efficiency for stakeholders**
- Designed and implemented **IAM roles** and **custom policies** to manage access control for AWS services, ensuring least-privilege access while maintaining operational efficiency
- Reduced **AWS costs by 20%** by identifying underutilized resources, enabling auto-scaling, and implementing S3 lifecycle policies to transition data to lower-cost storage tiers
- Configured **AWS Budgets** and **CloudWatch alerts** to track and manage cloud expenditure, ensuring proactive responses to cost anomalies
- Integrated **IAM roles** with data pipelines in **Glue**, **Redshift**, and **S3**, ensuring secure and automated access between services

Environment: Python, PowerBI, Tableau, AWS Glue, Kafka, Lambda, Redshift, S3, CloudWatch, IAM, Datadog, Pyspark, Airflow

SUNPOWER CORPORATION, RICHMOND, CA, JANUARY 2017 – JANUARY 2021

Senior Data Engineer

- Led REST API development for internal teams to access system energy performance data of millions of customers, incorporating business rules to ensure data quality. Built automated verification and validation tests, reducing testing time by 40%
- Designed and developed scalable ETL pipelines using AWS Glue, AWS Lambda, and S3, enabling seamless integration of data from diverse sources
- Reduced data pipeline execution time by 40% through performance optimization techniques, including partitioning, compression, and efficient data transformations in **AWS Glue**

- Implemented a centralized **S3-based and Redis data lake** to store structured and semi-structured data, improving data accessibility and enabling analytics use cases across teams
- Migrated data from legacy systems to **Amazon Redshift**, performing data modeling and schema optimization to support large-scale queries and reporting needs
- Automated data pipeline workflows using **AWS Step Functions** and **CloudWatch**, ensuring timely data ingestion, transformation, and loading with minimal manual intervention
- Developed robust **data validation** and monitoring frameworks using **Python** and AWS services to ensure data quality and integrity, minimizing data discrepancies
- Ensured data pipeline security and compliance by implementing encryption, access control policies, and AWS **Key Management Service (KMS)** for sensitive data
- Conducted cost analyses and optimized AWS resources by using lifecycle policies for S3 data storage and tuning Redshift clusters, resulting in a **20% cost reduction**
- Utilized Infrastructure as Code (IaC) tools such as **AWS CloudFormation** and **Terraform** to define and provision AWS infrastructure in a consistent and automated manner.
- Collaborated with business user's/product owners/developers to contribute to the analysis of functional requirements.
- Implemented **Spark SQL queries** that combine hive queries with Python programmatic data manipulations supported by RDDs and data frames.
- Designed and deployed **dynamic dashboards** that visualize sentiment trends, content categories, and key insights from customer reviews and comments for a user-friendly experience.
- Integrated data from diverse sources, including social media platforms, web scraping, and internal databases, for a holistic view of customer feedback and market trends, enhancing the accuracy and completeness of text analysis results.
- Collaborated with **DevOps** teams to automate software deployment and orchestration using **AWS CodeDeploy**, **Jenkins**, and containerization with **Docker** and **Amazon ECS**, ensuring efficient software delivery.

Environment: Python, Tableau, AWS Glue, Lambda, S3, Redshift, Step Functions, CloudWatch, KMS

SUNPOWER CORPORATION, RICHMOND, CA, AUGUST 2012 – JANUARY 2017

Data Engineer

- Developed custom **data preprocessing pipelines**, including **feature engineering and data augmentation**, to enhance the quality and diversity of input data for machine learning models
- Created and managed custom **JIRA** workflows, issue types, and project boards, tailored to the specific needs of data science and machine learning projects.
- Developed and organized **Confluence** pages for data science projects, including project charters, technical specifications, and project status reports, ensuring transparency and accessibility for all stakeholders.
- Implemented data streaming and real-time data processing using technologies like **Kafka** and **Apache Flink**, enhancing data pipeline efficiency and enabling real-time decision-making.
- Established an end-to-end **MLOps** pipeline, including version control, continuous integration, and automated deployment, using Git, Jenkins, and Kubernetes for containerization.

- Integrated monitoring and alerting systems like **Grafana** to proactively identify and address performance issues in deployed machine learning models.
- Built interactive dashboards using **Looker** and **Tableau** to monitor the real-time performance and key metrics of machine learning models, providing stakeholders with actionable insights and enabling data-driven decision-making for model optimization and fine-tuning.
- Successfully migrated applications to the cloud with Red Hat services and achieved a **15% monthly cost reduction**. Integrated robust **version control** through **Git** and establishing seamless **CI/CD** pipelines with **Jenkins**
- Developed comprehensive metrics and real-time dashboards to monitor the health of data streaming pipelines built using **Apache Kafka**, ensuring continuous data quality and performance optimization
- Leveraged **Snowflake** to optimize queries, resulting in a **20% improvement in data retrieval** speed and scalability for large healthcare datasets, enhancing data engineering and machine learning workflows
- **Utilized Power BI desktop to design real time dashboards and scorecards.** Implemented these with power pivot to help the client executive management to make decisions as per business requirements on accurate analytics data.
- Encoded and decoded **JSON** objects using **PySpark** to create and modify the dataframes in **Apache Spark**.
- Extensively used Stash **Git-Bucket** for Code Control and Worked on AWS Components such as **Airflow**, **Elastic Map Reduce (EMR)**, **Athena** and **Snowflake**.
- Container management using **Docker** by writing Docker files and setting up the automated build on Docker HUB and installing and configuring **Kubernetes**.
- Worked extensively with importing metadata into Hive and migrated existing tables and applications to work on **Hive** and **AWS cloud** and making the data available in **Athena** and **Snowflake**.
- Created **AWS Glue** job for archiving data from **Redshift** tables to **S3** (online to cold storage) as per data retention requirements and involved in managing S3 data layers and databases including Redshift and Postgres.
- Used AWS data pipeline for Data Extraction, Transformation and Loading from homogeneous or heterogeneous data sources and built various graphs for business decision making using Python **matplotlib** library
- Install and configure Apache Airflow for **S3 bucket** and Snowflake data warehouse and create **DAGs** to run the **Airflow**. Loaded the data into **Spark RDD** and performed in-memory data computation to generate the output response.
- Spearheaded the successful migration of legacy on-premises applications to the cloud with **Amazon Web Services (AWS)**, achieving a **20% reduction** in infrastructure costs and ensuring scalability and reliability.
- Managed and optimized database systems such as **MySQL**, **SQL Server**, and **MongoDB**, ensuring efficient data storage and retrieval, while minimizing downtime and enhancing overall system performance.
- Utilized database partitioning strategies to improve query performance for large datasets, resulting in a **30% reduction** in query response times.

Environment: PowerBI, Tableau, AWS cloud, EC2, RDS, S3, CloudWatch, CloudFront, Route 53, Microsoft Power BI, Amazon Redshift, AWS Lambda, Airflow, Spark RDD, Docker, Kubernetes, Snowflake